Scalable Step-Level Value Estimation for LLM Reasoning: 1-Step Fitted Backward Induction and Fitted Vine Interpolation

Technical Note

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Abstract

Reinforcement learning (RL) for multi-step large language model (LLM) reasoning requires assigning credit to intermediate reasoning steps when only sparse, terminal correctness feedback ($R \in \{0, 1\}$) is available. Standard Proximal Policy Optimization (PPO) trains a parametric value network (critic) on single-trajectory rollouts, which struggles due to high-variance binary returns and noisy single-sample bootstrapping. Conversely, critic-free rollout methods like VinePPO estimate state values via auxiliary Monte Carlo rollouts to termination from every step, incurring quadratic decoding cost $\mathcal{O}(K \cdot T^2)$ in the number of reasoning steps T.

This note presents two practical critic-assisted approaches that reduce the rollout cost of VinePPO while producing step-level value estimates $\hat{V}(s_t)$ for standard PPO on the base trajectory:

1. **Variant 1: 1-Step Fitted Backward Induction (FBI)**, which replaces full-horizon rollouts at early steps with 1-step branched Bellman expectations ($\mathcal{O}(K \cdot T)$ auxiliary steps), anchored by terminal Monte Carlo completions at the final step.
2. **Variant 2: Fitted Vine Interpolation and Subsampling**, which fits V_ϕ directly to unbiased $\hat{V}_{\mathrm{MC}}(s_t)$ targets (avoiding bootstrapping drift), tracks the critic's out-of-sample error $\sigma^2_{\mathrm{critic}}$ against $\hat{V}_{\mathrm{MC}}$ online, and uses precision-weighted shrinkage and checkpoint subsampling to reduce rollout cost to $p \cdot \mathcal{O}(K \cdot T^2)$.

1 Background and Motivation

1.1 Step-Level MDP for Reasoning

We formulate multi-step reasoning as a finite-horizon Markov Decision Process (MDP) over discrete reasoning steps (e.g., paragraphs or equations) rather than individual tokens. Given a prompt $x \sim \mathcal{D}$, an autoregressive policy π_θ generates a base trajectory of T steps:

$$\tau = (s_0, a_0, s_1, a_1, \ldots, s_{T-1}, a_{T-1}, s_T), \tag{1}$$

where $s_0 = x$, each action $a_t \sim \pi_\theta(\cdot \mid s_t)$ is a reasoning step, and state transitions are deterministic prefix concatenations $s_{t+1} = (s_t, a_t)$. At termination s_T, the environment provides a binary correctness reward $R(s_T) \in \{0, 1\}$, with intermediate rewards $r_t = 0$ for $t < T$. The true state value under policy π_θ is the probability of reaching a correct final answer from prefix s_t:

$$v^{\pi_\theta}(s_t) = \mathbb{E}_{\tau \sim \pi_\theta}[R(s_T) \mid s_t] \in [0, 1]. \tag{2}$$

1.2 The Trade-Off Between Standard PPO and VinePPO

To compute step-level advantages $\hat{A}(s_t, a_t) = \hat{V}(s_{t+1}) - \hat{V}(s_t)$ along the base trajectory τ, standard PPO and VinePPO sit at opposite extremes of the compute–variance trade-off:

1. **Standard PPO Critic ($\mathcal{O}(T)$ rollout cost):** Fits a parametric critic $V_\phi(s_t)$ using only the single sampled trajectory τ. With $\lambda = 1$ (Monte Carlo), each state regresses against a single binary outcome $R(s_T) \in \{0, 1\}$, which has maximal Bernoulli variance $v(1-v) \leq 0.25$. With $\lambda < 1$ (TD bootstrapping), the target $V_\phi(s_{t+1})$ relies on a single noisy action realization a_t and can drift when downstream value estimates are inaccurate.

2. **VinePPO ($\mathcal{O}(K \cdot T^2)$ rollout cost):** Bypasses a parametric critic entirely. At every intermediate state s_t, it samples K independent rollouts all the way to termination and averages their terminal rewards:

$$\hat{V}_{\mathrm{MC}}(s_t) = \frac{1}{K} \sum_{k=1}^{K} R\Big(\tau^{(k)}_{t \to \mathrm{term}}\Big). \tag{3}$$

   While $\hat{V}_{\mathrm{MC}}(s_t)$ is unbiased ($\mathbb{E}[\hat{V}_{\mathrm{MC}}(s_t)] = v^{\pi_\theta}(s_t)$) with sampling variance $\frac{v(1-v)}{K}$, running K full completions from each of the T steps requires $\sum_{t=0}^{T-1} K(T-t) = K\frac{T(T+1)}{2} = \mathcal{O}(K \cdot T^2)$ auxiliary steps.

In both variants below, we follow VinePPO in using auxiliary rollouts strictly to obtain step-level value estimates $\hat{V}(s_t)$ along the base trajectory τ. The policy π_θ is then updated on the base trajectory using the standard clipped PPO objective:

$$\mathcal{L}_{\mathrm{PPO}}(\theta) = \mathbb{E}_\tau \left[\sum_{t=0}^{T-1} \min\Big(\rho_t(\theta)\hat{A}(s_t, a_t),\ \mathrm{clip}(\rho_t(\theta), 1-\epsilon_{\mathrm{clip}}, 1+\epsilon_{\mathrm{clip}})\hat{A}(s_t, a_t) \Big) \right], \tag{4}$$

where $\hat{A}(s_t, a_t) = \hat{V}(s_{t+1}) - \hat{V}(s_t)$ (or GAE over step TD residuals) and $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\mathrm{old}}(a_t|s_t)}$.

2 Variant 1: 1-Step Fitted Backward Induction (FBI)

Variant 1 replaces full-horizon auxiliary rollouts at early steps with 1-step branched lookahead backed by a learned critic V_ϕ, reducing auxiliary decoding complexity to $\mathcal{O}(K \cdot T)$.

2.1 1-Step Lookahead Targets ($K' = 1$)

Along a base trajectory $\tau = (s_0, a_0, \ldots, s_T)$, we construct targets $y(s_t)$ for each state s_t:

- **Terminal Boundary ($t = T$):** The completed base trajectory has ground-truth value $y(s_T) = R(s_T) \in \{0, 1\}$.

- **Monte Carlo Anchor ($t = T-1$):** At the penultimate state s_{T-1}, rollouts to completion are short. We sample K completions $\tau^{(1..K)}_{T-1 \to \mathrm{term}} \sim \pi_\theta(\cdot \mid s_{T-1})$ to termination and compute the Monte Carlo average:

$$y(s_{T-1}) = \frac{1}{K} \sum_{k=1}^{K} R\Big(\tau^{(k)}_{T-1 \to \mathrm{term}}\Big). \tag{5}$$

- **1-Step Inductive Lookahead ($t < T-1$):** At earlier states, we sample K *single-step* continuations $a_t^{(1..K)} \sim \pi_\theta(\cdot \mid s_t)$, yielding next states $s_{t+1}^{(k)} = (s_t, a_t^{(k)})$, and average their values:

$$y(s_t) = \frac{1}{K} \sum_{k=1}^{K} v_{\mathrm{next}}\Big(s_{t+1}^{(k)}\Big), \quad \text{where } v_{\mathrm{next}}(s') = \begin{cases} R(s') & \text{if } s' \text{ is terminal,} \\ \mathrm{sg}(V_\phi(s')) & \text{otherwise.} \end{cases} \tag{6}$$

Averaging over K single-step branches in eq. (6) approximates the expected Bellman backup $\mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}[V_\phi(s_{t+1})]$ rather than bootstrapping from a single action realization.

2.2 Interleaved Backward Fitting with Replay Anchoring

To ensure backward induction propagates fresh value targets from $t = T-1$ toward $t = 0$ within an iteration:

1. **Just-in-Time Target Evaluation:** We iterate backward from $\Delta = 1$ to $T_{\max}$ ($t = T-\Delta$). At step t, $y(s_t)$ is computed using the *just-updated* critic weights ϕ from step $t+1$.
2. **Cumulative Replay Anchoring:** We maintain a cumulative dataset $\mathcal{D}_{\text{fit}}$ of already-resolved pairs $(s_{t'}, y(s_{t'}))$ for $t' \geq t$ (including terminal branch endpoints) during the backward sweep so gradient updates at early steps $t \ll T$ do not overwrite the terminal anchor learned at $t = T-1$.

3 Variant 2: Fitted Vine Interpolation and Subsampling ($V_\phi \leftrightarrow \hat{V}_{\text{MC}}$)

A key limitation of Variant 1 is that errors in V_ϕ on off-trunk sibling branches $s_{t+1}^{(k)}$ still compound over $T-1$ bootstrapping steps. Variant 2 eliminates multi-step bootstrapping altogether by fitting $V_\phi(s_t)$ **directly to unbiased VinePPO Monte Carlo targets** $\hat{V}_{\text{MC}}(s_t)$, and leveraging an online measurement of V_ϕ's accuracy relative to $\hat{V}_{\text{MC}}$ to reduce variance and rollout compute.

3.1 Direct Supervised Distillation and Online Accuracy Auditing

Suppose we evaluate K-rollout Monte Carlo targets $\hat{V}_{\text{MC}}(s_t) = \frac{1}{K}\sum_{k=1}^{K} R(\tau_{t\to\text{term}}^{(k)})$ on a set of states $\mathcal{S}_{\text{MC}}$. Because $\hat{V}_{\text{MC}}(s_t)$ is an average of K i.i.d. Bernoulli trials, its unbiased per-state sampling variance estimate is:

$$\hat{\sigma}^2_{\text{MC}}(s_t) = \frac{\hat{V}_{\text{MC}}(s_t)\big(1-\hat{V}_{\text{MC}}(s_t)\big)}{K-1} \quad (\text{or } \tfrac{1}{4K} \text{ conservative bound for } K=1). \tag{7}$$

Before updating the critic V_ϕ on the current batch $\mathcal{S}_{\text{MC}}$, we audit the critic's out-of-sample mean squared error against $\hat{V}_{\text{MC}}$ and isolate the critic's excess prediction error variance σ^2_{critic} via an exponential moving average (EMA):

$$\hat{\epsilon}^2_{\text{batch}} = \frac{1}{|\mathcal{S}_{\text{MC}}|}\sum_{s_t\in\mathcal{S}_{\text{MC}}} \max\Big(0,\ \big(V_\phi(s_t) - \hat{V}_{\text{MC}}(s_t)\big)^2 - \hat{\sigma}^2_{\text{MC}}(s_t)\Big), \quad \sigma^2_{\text{critic}} \leftarrow (1-\beta)\sigma^2_{\text{critic}} + \beta\hat{\epsilon}^2_{\text{batch}}. \tag{8}$$

After recording the out-of-sample audit σ^2_{critic} and computing the interpolated values below, we fit V_ϕ directly to the soft probability targets $\hat{V}_{\text{MC}}(s_t) \in [0,1]$ by minimizing $\mathbb{E}_{s_t\in\mathcal{S}_{\text{MC}}}\Big[\big(V_\phi(s_t) - \hat{V}_{\text{MC}}(s_t)\big)^2\Big]$.

3.2 Mechanism A: Precision-Weighted Shrinkage Interpolation (Small K)

When K is kept small (e.g., $K = 2$) to save rollout compute, $\hat{V}_{\text{MC}}(s_t)$ has noticeable binomial sampling variance $\hat{\sigma}^2_{\text{MC}}(s_t)$, whereas $V_\phi(s_t)$ has zero rollout sampling variance but carries prediction error variance σ^2_{critic}. The minimum-MSE linear combination (Empirical Bayes shrinkage) weights each estimator inversely to its variance:

$$\hat{V}_{\text{interp}}(s_t) = (1-\alpha_t)\,\hat{V}_{\text{MC}}(s_t) + \alpha_t\, V_\phi(s_t), \qquad \alpha_t = \frac{\hat{\sigma}^2_{\text{MC}}(s_t)}{\hat{\sigma}^2_{\text{MC}}(s_t) + \sigma^2_{\text{critic}} + \epsilon}. \tag{9}$$

- **Cold-Start Safety ($\sigma^2_{\text{critic}} \gg \hat{\sigma}^2_{\text{MC}}$):** Early in training when V_ϕ is inaccurate, $\alpha_t \approx 0$ and $\hat{V}_{\text{interp}}(s_t)$ reduces to pure VinePPO ($\hat{V}_{\text{MC}}(s_t)$).
- **Variance Reduction ($\sigma^2_{\text{critic}} \lesssim \hat{\sigma}^2_{\text{MC}}$):** As V_ϕ accurately tracks $\hat{V}_{\text{MC}}$, α_t increases, shrinking noisy small-K Monte Carlo estimates toward $V_\phi(s_t)$.

3.3 Mechanism B: Subsampled Vine Rollouts ($p \cdot \mathcal{O}(K \cdot T^2)$ Cost)

With a calibrated critic V_ϕ trained on $\hat{V}_{\text{MC}}$, we do not need to run K full rollouts at every intermediate step:

1. For each intermediate step s_t ($t < T$), we flip a coin Bernoulli(p) with subsample rate $p \in (0, 1]$ (e.g., $p = 0.25$).
2. **On sampled checkpoints ($s_t \in \mathcal{S}_{\text{MC}}$):** We run K full rollouts to compute $\hat{V}_{\text{MC}}(s_t)$, audit σ^2_{critic} via eq. (8), and set $\hat{V}(s_t) = \hat{V}_{\text{interp}}(s_t)$ via eq. (9).
3. **On unsampled checkpoints ($s_t \notin \mathcal{S}_{\text{MC}}$):** We skip auxiliary rollouts and use the critic prediction $\hat{V}(s_t) = V_\phi(s_t)$ directly (or fall back to rollout generation if σ^2_{critic} exceeds a trust threshold τ_{err}).

4 Comparison and Trade-offs

Method	Auxiliary Steps / Traj.	Complexity	Bootstrapping Drift?	Critic Required?
Standard PPO (GAE)	0 (T base)	$\mathcal{O}(T)$	High ($\lambda < 1$)	Yes
VinePPO	$K \cdot \frac{T(T+1)}{2}$	$\mathcal{O}(K \cdot T^2)$	None (Unbiased MC)	No
Variant 1: 1-Step FBI	$\approx K \cdot T$	$\mathcal{O}(K \cdot T)$	Moderate ($T-1$ steps)	Yes
Variant 2: Fitted Vine	$p \cdot K \frac{T(T+1)}{2}$	$p \cdot \mathcal{O}(K \cdot T^2)$	**None (Direct $\hat{V}_{\text{MC}}$ fit)**	Yes

Table 1: **Comparison of step-level value estimation methods.** Variant 1 achieves linear $\mathcal{O}(K \cdot T)$ scaling via 1-step dynamic programming; Variant 2 avoids bootstrapping drift entirely by regressing V_ϕ on subsampled $\hat{V}_{\text{MC}}$ targets and interpolating via measured critic accuracy.

- **When Variant 1 (1-Step FBI) is preferable:** When T is large enough that full completions from early steps are prohibitively long, but T is still small enough ($T \approx 4$–8 chunks) that $T - 1$ stages of 1-step bootstrapping do not severely compound off-trunk critic errors.
- **When Variant 2 (Fitted Vine Interpolation) is preferable:** When avoiding TD bootstrapping instability and OOD sibling queries is paramount. Because Variant 2 supervises V_ϕ directly on unbiased $\hat{V}_{\text{MC}}(s_t)$ values and gates its contribution via out-of-sample error σ^2_{critic}, a poorly fitted critic automatically receives near-zero weight ($\alpha_t \to 0$) rather than corrupting upstream targets.

5 Algorithm Summaries

Algorithm 1 Variant 1: 1-Step Fitted Backward Induction PPO (FBI-PPO)

Require: Policy π_θ, Critic V_ϕ, prompts $\mathcal{D}$, batch size B, branch factor K.

for each training iteration **do**
  Sample B prompts $\{x_i\}_{i=1}^{B} \sim \mathcal{D}$, generate base trajectories $\{\tau_i\}_{i=1}^{B} \sim \pi_\theta$, and set $\mathcal{D}_{\text{fit}} \leftarrow \emptyset$.
  for each base trajectory $\tau = (s_0, a_0, \ldots, s_T)$ in batch **do**
    Add terminal state $(s_T, R(s_T))$ to $\mathcal{D}_{\text{fit}}$.
    At $t = T-1$**:** Sample K completions $\tau_{T-1\to\text{term}}^{(1..K)} \sim \pi_\theta(\cdot \mid s_{T-1})$ with rewards $R^{(1..K)}$.
    At $t < T-1$**:** Sample K single steps $a_t^{(1..K)} \sim \pi_\theta(\cdot \mid s_t)$ to form $s_{t+1}^{(k)} = (s_t, a_t^{(k)})$.
  end for
  Add penultimate anchors $\big(s_{T-1},\, y(s_{T-1}) = \frac{1}{K}\sum_{k=1}^{K} R^{(k)}\big)$ and terminal branch endpoints to $\mathcal{D}_{\text{fit}}$.
  Update ϕ by minimizing $\mathbb{E}_{(s,y)\sim\mathcal{D}_{\text{fit}}}\big[(V_\phi(s) - y)^2\big]$.
  for remaining step distance $\Delta = 2$ up to $T_{\max}$ (i.e., $t = T - \Delta$ moving backward) **do**
    for each trajectory in batch with step index $t = T - \Delta \geq 0$ **do**
      $y(s_t) \leftarrow \frac{1}{K}\sum_{k=1}^{K} v_{\text{next}}\Big(s_{t+1}^{(k)}\Big)$, where $v_{\text{next}}(s') = R(s')$ if terminal else $\text{sg}(V_\phi(s'))$.
      Append $(s_t, y(s_t))$ to cumulative buffer $\mathcal{D}_{\text{fit}}$.
    end for
    Update critic ϕ on $\mathcal{D}_{\text{fit}}$ via gradient descent on $\mathbb{E}_{(s,y)\sim\mathcal{D}_{\text{fit}}}\big[(V_\phi(s) - y)^2\big]$.
  end for
  Set $\hat{V}(s_t) \leftarrow y(s_t)$ for $t < T$ and $\hat{V}(s_T) \leftarrow R(s_T)$; update θ on $\{\tau_i\}_{i=1}^{B}$ via PPO (eq. (4)).
end for

Algorithm 2 Variant 2: Fitted Vine Interpolation & Subsampled PPO

Require: Policy π_θ, Critic V_ϕ, prompts $\mathcal{D}$, batch size B, rollouts K, subsample rate $p \in (0, 1]$, EMA rate β.

for each training iteration **do**
  Sample B prompts $\{x_i\}_{i=1}^B \sim \mathcal{D}$, generate base trajectories $\{\tau_i\}_{i=1}^B \sim \pi_\theta$, and set $\mathcal{S}_{\text{MC}} \leftarrow \emptyset$.
  for each base trajectory $\tau = (s_0, a_0, \ldots, s_T)$ and step $t \in \{0, \ldots, T-1\}$ **do**
    if Bernoulli$(p) == 1$ **then**
      Sample K full rollouts $\tau_{t\to\text{term}}^{(1..K)} \sim \pi_\theta(\cdot \mid s_t)$ and compute $\hat{V}_{\text{MC}}(s_t) \leftarrow \frac{1}{K}\sum_{k=1}^K R(\tau_{t\to\text{term}}^{(k)})$.
      Compute MC variance $\hat{\sigma}^2_{\text{MC}}(s_t)$ via eq. (7) and add $(s_t, \hat{V}_{\text{MC}}(s_t), \hat{\sigma}^2_{\text{MC}}(s_t))$ to $\mathcal{S}_{\text{MC}}$.
    end if
  end for
  Audit Critic Out-of-Sample: Update σ^2_{critic} on $\mathcal{S}_{\text{MC}}$ using current V_ϕ via eq. (8).
  for each base trajectory $\tau = (s_0, a_0, \ldots, s_T)$ **do**
    Set $\hat{V}(s_T) \leftarrow R(s_T)$.
    for $t = 0$ to $T-1$ **do**
      if $s_t \in \mathcal{S}_{\text{MC}}$ **then**
        $\alpha_t \leftarrow \frac{\hat{\sigma}^2_{\text{MC}}(s_t)}{\hat{\sigma}^2_{\text{MC}}(s_t)+\sigma^2_{\text{critic}}+\epsilon}$; $\quad \hat{V}(s_t) \leftarrow (1-\alpha_t)\hat{V}_{\text{MC}}(s_t) + \alpha_t V_\phi(s_t)$.
      else
        $\hat{V}(s_t) \leftarrow V_\phi(s_t)$.
      end if
    end for
  end for
  Fit Critic to $\hat{V}_{\text{MC}}$: Update ϕ by minimizing $\mathbb{E}_{(s_t, \hat{V}_{\text{MC}})\sim\mathcal{S}_{\text{MC}}}\left[(V_\phi(s_t) - \hat{V}_{\text{MC}}(s_t))^2\right]$ (plus terminal $(s_T, R(s_T))$).
  Update policy parameters θ on base trajectories $\{\tau_i\}_{i=1}^B$ via PPO (eq. (4)) using $\hat{A}(s_t, a_t) = \hat{V}(s_{t+1}) - \hat{V}(s_t)$.
end for